



How to get children moving? The effectiveness of school-based interventions promoting physical activity in children and adolescents – A systematic review and meta-analysis of randomized controlled- and controlled studies

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ABSTRACT

This systematic review and meta-analysis summarizes the effectiveness of school-based physical activity (PA) interventions on children's and adolescents' PA. As no robust empirical evidence exists regarding what seems to characterize the school-based PA interventions that are most effective, we further aimed to map key factors of particular importance when trying to increase PA in early stages of life through school-based strategies.

Intervention effects were calculated as standardized between-group (i.e., intervention vs. control) mean differences (SMD) in PA from baseline to follow-up. In total, 189 publications were included. Few studies (7%) were of high quality. Our results demonstrate that school-based interventions only have a small positive effect on children's and adolescents' PA levels. Compared to the effect observed during total day (SMD = 0.27, $p < 0.001$), a slightly larger effect was observed during school hours (SMD = 0.37, $p < 0.001$), while no intervention effect was observed during leisure time (SMD = 0.07, $p = 0.20$). There was a tendency for interventions to be more effective if theoretical frameworks for behavior changes were used in the design phase. The largest effect size was observed when experts from outside school delivered the program (SMD = 0.56, $p = 0.01$), but training of personnel involved in delivery was the determining factor for program effectiveness as no effect was observed if interventions were delivered primarily by schools' untrained staff (SMD = 0.06, $p = 0.61$). Intervention effects were larger if parents were involved in the intervention program (parents involved: SMD = 0.35, $p < 0.001$; parents not involved: SMD = 0.16, $p = 0.02$). Small positive intervention effects were sustained at long-term follow-up after end of intervention. Overall, the certainty of the evidence of the findings is rated as low.

1. Introduction

Physical inactivity has major negative health effects accounting for

more than 5 million yearly deaths worldwide (Lee et al., 2012a). Lack of physical activity (PA), and the consequences hereof, including reduced aerobic fitness and excessive body weight, adversely affect health not

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only in adults but also in children (Andersen et al., 2006; Resaland et al., 2010; Barton, 2012). Moreover, PA levels and correlated risk factors during childhood and adolescence track into adulthood (Bao et al., 1994; Twisk et al., 1997; Joshi et al., 2014) and are associated with cardiovascular events later in life (Joshi et al., 2014). In addition, insufficient PA and overweight negatively influence quality of life and self-efficacy (Anokye et al., 2012; Danielsen et al., 2012). Also, the contemporary consensus is that children's and adolescents' PA is important for cognitive function, engagement and motivation, physiological well-being, as well as social inclusion (Bangsbo et al., 2016).

Despite the well-known health benefits of PA, a large proportion of children worldwide do not meet PA recommendations (Aubert et al., 2022; Ekelund et al., 2012). Children and adolescents are, therefore, a relevant target group for early promotion of PA. The school setting is widely regarded as an ideal arena for fostering change in health-related behaviors, as it e.g., facilitates close interaction with teachers and students within an educational context favorable for promoting health (Pate et al., 2006). As the vast majority of all children and adolescents spend a considerable portion of their waking hours at school, health-promoting interventions implemented within the school setting also have the potential to reach all pupils, irrespective of background and socio-economic status (Kriemler et al., 2011). As an example, Danish children accumulate only 28 and 8 min of daily moderate-to-vigorous PA (MVPA) during school time and physical education lessons, respectively (Pedersen et al., 2020), suggesting much potential for increasing PA in schools.

Findings regarding the effects of school-based PA interventions on children's and adolescents' level of PA are inconsistent, but meta-analyses have suggested that school-based PA interventions seem to have no (Love et al., 2019) or only minor positive effects (Metcalfe et al., 2012; Hynynen et al., 2016; Nally et al., 2021; Neil-Sztramko et al., 2021). Some systematic reviews suggest that the most effective interventions aimed at increasing children's PA level include different forms of PA-promoting school initiatives combined with various efforts in the local community (Kriemler et al., 2011; van Sluijs et al., 2007). However, school-based PA interventions are often complex multi-component programs, implemented in various school contexts. From a recent meta-analysis that summarized the impact of school-based interventions on children's obesity related behavior it was not possible to determine which components of the interventions were most effective (Nally et al., 2021). Consequently, it is unknown which school-based strategies are most effective in promoting healthy lifestyles in children and adolescents.

We need more knowledge regarding what characterizes the most effective PA interventions undertaken at schools. Therefore, in light of the complexity and diversity of many school-based interventions, the ambition in the present study was to identify and highlight some of the overarching sources of variation observed in intervention effectiveness. More specifically, our aims were 1) to summarize the evidence regarding the effect of school-based PA interventions on children's and adolescents' PA levels across various everyday domains, and 2) to identify key factors explaining the variation in effectiveness of school-based PA interventions on children's and adolescents PA levels. Ultimately, our ambition is to provide evidence-based conclusions that can inform practice, whether in adjusting existing PA programs or in the development of new initiatives.

2. Methods and methods

This systematic review and meta-analysis is reported in accordance with the PRISMA (Preferred Reporting Items for Systematic Reviews and Meta-Analyses) guidelines and the protocol was registered in the PROSPERO international register of systematic reviews PROSPERO (Registration number: CRD42014013570). Since the original registration, we excluded sedentary behavior as an outcome and used a customized version of the risk of bias tool to increase clarity and

relevance.

2.1. Literature search

We searched 10 different electronic databases (Pubmed, Embase, Scopus, Web of Science, Cochrane CENTRAL, Physical Education Index, Academic Search Premier, Cinahl, SOC-index, SportsDiscus) for scientific literature in which the effect of school-based interventions on children's PA level was examined. We also searched the SIGLE database for grey literature in the form of e.g., research reports and dissertations.

A research librarian was consulted during selection, combination, and testing of search terms. The search strategy was based on a search-matrix consisting of four key elements: 1) population (e.g., child, adolescent), 2) setting (e.g., school, classroom), 3) study design (e.g., clinical controlled trial, intervention), 4) intervention (e.g., physical activity, exercise) adapted to the individual databases (see [S1 Appendix](#) for the complete search strategies adapted to the various databases used).

We conducted the first search in November of 2014 (phase 1). Search results were limited to publications published after 2006 for two reasons. First, to enhance generalizability of findings to current school and societal structures and environment. Secondly, because a systematic review by Dobbins et al. investigating the effectiveness of school-based interventions promoting PA and fitness in children and adolescents aged 6–18 years had been conducted including studies published up to year 2007 (Dobbins et al., 2009). The findings based on the literature search conducted in phase 1 have been outlined in a Danish report aimed for Danish organizations and municipalities (Møller et al., 2019). To include trials published after November 2014, the literature search was updated in August 2020 (phase 2). This involved performing an “upward citation search”, meaning that all publications in which one of the studies included in phase 1 (i.e., 112 studies) were cited were further checked against the inclusion and exclusion criteria.

2.2. Study selection

After performing the literature search the retrieved publications were checked for eligibility on title/abstract level independently by two reviewers in groups of two (NCM and MGBR or LØ and a research librarian. We then obtained full texts for studies that we judged potentially eligible. Disagreements within groups were resolved through discussion until the group reached a consensus. Identical methodology was applied in the phase 2-literature search, except this time retrieved publications were checked independently by NCM and KTL.

2.3. Inclusion and exclusion criteria

Studies were deemed eligible if they 1) targeted school children or adolescents aged 6–18 years (referred to as either “children and adolescents” or “children”) at onset of the intervention; 2) examined the effect of a school-based intervention program that incorporated a PA promoting component; 3) compared results observed in the intervention group with results observed in a control group receiving no PA intervention beyond the standard physical education curriculum; 4) reported children's PA by means of total PA, MVPA, moderate-, or vigorous PA during a full day, leisure time or school time; 5) included assessments performed in the same participants both before start of the intervention and at least 8 weeks after; 6) were randomized controlled- or controlled trials (cluster or individual); 7) were published after 2006.

Studies were excluded if 1) the intervention was initiated exclusively for children diagnosed with a specific disease, or disability, or for children with a certain health problem (e.g., overweight, or high blood pressure); 2) the intervention was not primarily school-based (e.g., predominantly targeting children outside the school setting in community or local sports club). A full list of included studies is accessible in [S2 Appendix](#).

2.4. Data extraction

Two reviewers (NCM and MGBR) independently extracted information on study level covariates and results. Firstly, basic information, including authors, title, place, and year of publication was extracted. Secondly, we extracted data on children's age, sex, type and length of intervention, follow-up time reported in weeks, participant- and school sample size (N), PA assessment method and domain, time of follow-up measures, and effect size together with p-value, variance, or 95% confidence interval. When a trial both reported device-based PA results and PA results obtained by use of self-reports, only device-based results were extracted.

We also extracted information on study design, randomization procedure (in case of RCT), blinding status of study assessors, intervention- and control group comparability at baseline, and to which degree potential confounding was handled when groups appeared different at baseline.

Based on the a priori assumption that the intervention effect may be modified by how the intervention is fundamentally planned and designed, and ultimately delivered to the children, information was extracted from all studies pertaining to the following aspects: theoretical framework, intervention components, parental involvement, and program delivery.

2.5. Overall data analysis strategy

A primary meta-analysis was conducted to evaluate the short-term effect of school-based intervention programs on children's total daily PA, leisure-time PA, and school-time PA, respectively. This was done to obtain more detailed evidence on the effectiveness of school-based interventions on children's and adolescents' PA levels, as compensation e.g., may occur when attempting to increase children's physical activity levels through intervention participation (Beck et al., 2022). We used baseline and follow-up time closest to intervention endpoint to describe short-term intervention effects. We tested if a priori selected intervention characteristics moderated the intervention effect on children's and adolescents' PA (i.e., reduced between study variance Tau^2) by use of total daily PA since an indicator of a person's total PA typically is most determinative of most possible associations with health, and since the majority of included studies reported changes in children's total day PA. Hence, we tested the potential influence of the following intervention characteristics by using subgroups meta-analyses: 1) utilization of underpinning behavioral change theory (yes or no); 2) complexity of intervention (uni-component program focusing exclusively on PA, or multi-component program focusing on both PA and other health-related behaviors such as e.g., diet); 3) program delivery (experts from outside school, experts from outside school together with school staff, trained school staff, or untrained school staff); 4) parental involvement (yes or no). In a secondary analysis, we also examined the long-term post-intervention effect of school-based interventions on children's PA by using the follow-up time most distant from the completion of intervention.

Further, we tested in weighted random effect meta-regression models (Metareg in Stata 18.0) if there was a linear relation between intervention duration, or children's age, and overall summary effect sizes. Finally, sensitivity analyses were performed to explore if the PA construct (total PA vs. MVPA), method used to assess PA (self-reports vs. device-based), school stage (primary vs. secondary), study design (non-RCT vs. RCT), overall study quality (risk of bias summary score), and literature study-phase (phase 1 vs. phase 2) moderated the intervention effect size.

2.6. Calculating effect sizes

Various types of instruments were used to quantify PA across trials and results were reported in different units. Therefore, intervention

effects were calculated using Hedges g , as standardized between group (i.e., intervention vs. control) differences in mean PA change from baseline to follow-up, or in mean differences in PA at follow-up controlled for baseline PA, divided by the pooled standard deviation of PA changes for the intervention and control group. Standard deviations were calculated from 95% confidence intervals or p-values if they were not reported. We have previously observed stability coefficients up to 0.48 and 0.53 for total habitual PA over a 6-year period in boys and girls, respectively (Kristensen et al., 2008). However, when a trial reported PA means and standard deviations within groups at baseline and follow-up, respectively, we assumed a correlation of $r = 0.6$ between baseline and follow-up PA to estimate the standard deviation of mean PA change. This assumption was made due to the restriction of PA assessments to specific domains, and the fact that the follow-up period was typically considerably shorter than 6 years. When studies reported their findings in the form of odds ratios, or proportions, transformation of data to standardized means were performed using the formula: $SMD = \ln(OR)/1.81$ (Chinn, 2000). When multiple intervention effects were reported (e.g., for girls and boys, for different age-groups, or for subgroups receiving different intervention programs) we calculated a standardized mean difference for each group separately. In the case of e.g., two subgroups receiving different interventions, we divided the control group sample size by two, thereby avoiding falsely high study-weights due to control subjects being included multiple times in the same meta-analysis. Whenever a study reported several intervention effects all observed in the same domain (e.g., leisure time PA reported both in the morning before school and during after school hours) the effect size representing the largest proportion of time was included. Also, in the case where studies reported both changes in children's total PA and MVPA, only the effect size for total PA was included in the meta-analysis. In a very few cases where no total PA or MVPA outcomes were reported, vigorous or moderate PA intervention effects were estimated.

Overall summary effect size with 95% CI was calculated based on published data by using a restricted maximum likelihood random effect model in Stata version 18.0 as large heterogeneity was expected due to differences in e.g., intervention duration, PA promoting components, and school contexts. An overall standardized mean difference of ≈ 0.2 is considered as a small difference, ≈ 0.5 is considered as moderate, and ≈ 0.8 as a large difference (Cohen, 1988). Heterogeneity observed in individual study effects was formally quantified by using I^2 as a measure of the percentage of observed variability explainable by true between-study variations rather than by random variation. The following guide for I^2 interpretation has been suggested: 0%–40%: might not be important, 30%–60%: may represent moderate heterogeneity, 50%–90%: may represent substantial heterogeneity, 75%–100%: considerable heterogeneity (Higgins et al., 2023).

Publication bias was evaluated visually based on funnel plots and by applying Egger's test to check for funnel plot asymmetry. We used the post-command "trim" function in Stata 18 to simulate the summary effect size by eliminating small-study bias.

2.7. Assessing the quality of the included studies and certainty of evidence

Two authors (NCM and JSP) independently assessed the risk of bias of all included studies based on a customized tool inspired by the Effective Public Health Practice Project (EPHPP) quality assessment tool for quantitative studies (Thomas et al., 2004). Study quality was assessed based on five different items: 1) selection bias (i.e., allocation concealment secured through proper randomization procedure); 2) information bias (i.e., blinding of personnel assessing study outcome); 3) attrition bias (i.e., less than 20% drop-out); 4) confounding (i.e., intervention and control groups comparable at baseline); 5) the validity of method used to assess study outcome (i.e., device assessed PA vs. self-reported PA). In the case of doubt or disagreement consensus was reached through discussion. Studies were assigned a score (i.e., 0 points

in case of unclear or high risk of bias, and 1 point in case of low risk of bias) within each item. Scores were then summed, and all studies were labeled with a final quality score (i.e., 0–1: high risk of bias/low quality; 2–3: moderate risk of bias/moderate quality; 4–5: low risk of bias/high quality). For more information on assessing risk of bias, please see [S3 Appendix](#). We used the GRADE approach to judge the overall certainty of the evidence for all primary and secondary outcome measures as either very low, low, moderate, or high ([Guyatt et al., 2008](#)), as each item (i.e., risk of bias, inconsistency of results, indirectness, publication

bias, and imprecision) was considered to determine whether downgrading the level of certainty was needed.

3. Results

3.1. Inclusion and exclusion of studies

Fig. 1 illustrates the steps involved in identification, screening, exclusion, and inclusion of studies. Phase one yielded a total of 49,159

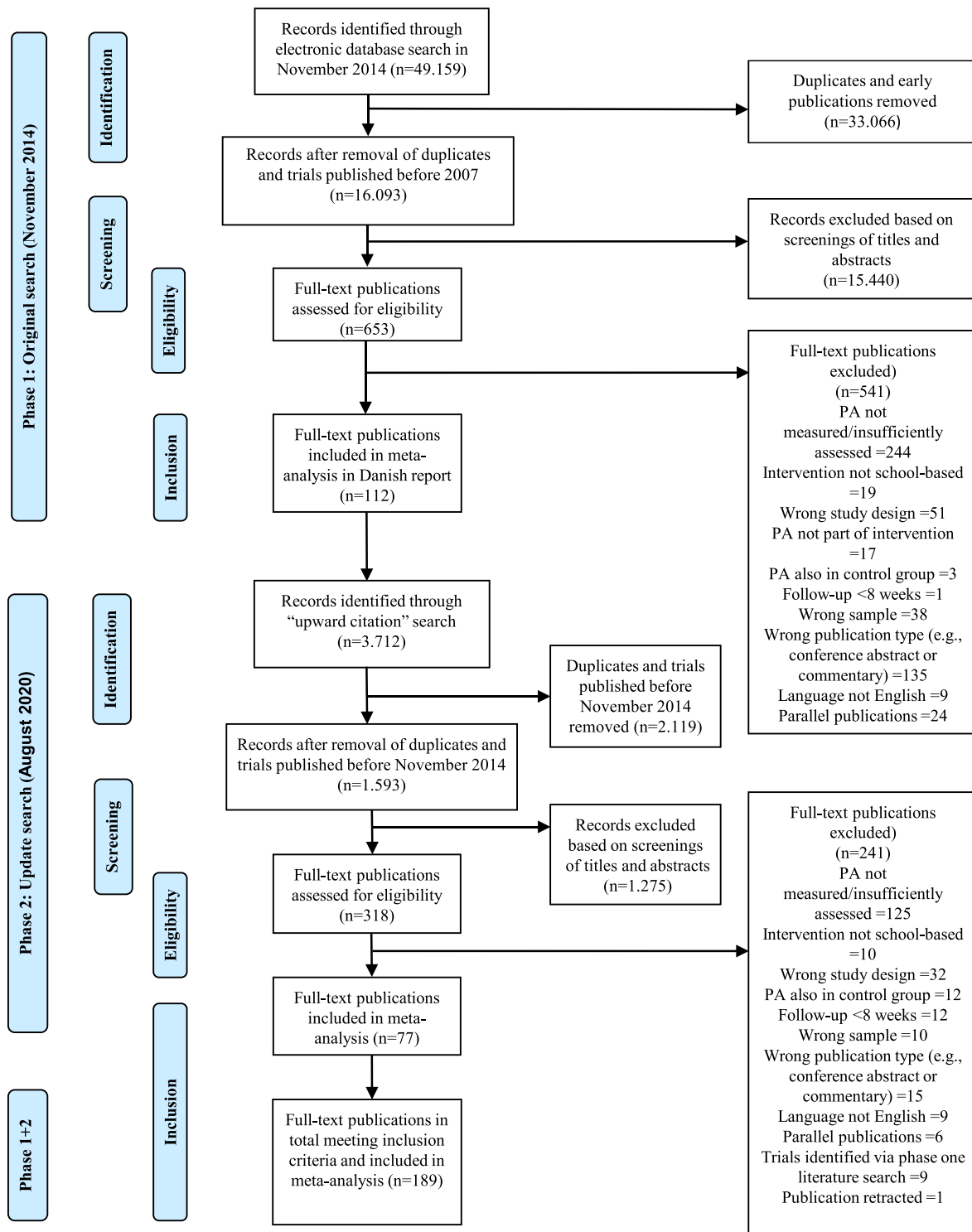


Fig. 1. Flowchart of study selection.

reports. After removing duplicates, and excluding trials published before year 2007, 16,093 records were retained. After title/abstract screening, we further excluded 15,440 publications. Of the 653 original articles, for which full text were received, further 541 were excluded. Consequently, 112 articles were deemed eligible for final inclusion. In the phase 2 literature search, 3217 records were identified. After removal of duplicates and studies published before November 2014, 1593 publications were retained and screened. We excluded 1275 articles based on title/abstract screening. After 318 full-text readings 77 articles were deemed eligible for inclusion. In total 189 studies, reporting the results of 179 unique school-based interventions, were included in the meta-analyses.

3.2. Description of school-based intervention programs and other study characteristics

This meta-analysis includes results of 44 trials implemented in US/Canada, 35 in Great Britain, 25 in Africa/Asia, 23 in Southern Europe, 22 in Australia/New Zealand, 19 in Western Europe, 11 in Scandinavia, 7 in South America/The Caribbean, and 3 implemented in Eastern Europe. One-hundred and thirty-three (70%) trials were RCTs of which 124 were cluster-RCTs. Intervention programs varied substantially regarding e.g., specific school context, content, and duration, but often they included various PA promoting components incorporated either during classroom lessons, physical education lessons, or recess. Program duration ranged from just a single classroom session to 4 years (mean duration: 42 weeks). Ninety-six studies reported results from uni-component interventions aimed solely to increase PA, while 93 studies reported results from multi-component intervention (i.e., interventions designed to increase both PA and other health-related behavior such as increased fruit intake or reduced intake of sweetened beverage). For a more detailed and complete description of all interventions, please see [S1 Table & S2 Table](#).

One-hundred and seventy-five publications (93%) reported both girls' and boys' intervention effects, either in the form of combined or gender-specific results, while only girls' results were described in 14 studies (7%).

Most subjects were aged 8–12 years (i.e., median (IQR) age = 10.0 (8.6–12) years). Various techniques were used to quantify their PA; subjective methods e.g., in the form of questionnaires or diaries were applied in 84 studies (44%), while devised-based assessments, typically accelerometers or more simple pedometers, were utilized in 102 studies (54%). Three studies (2%) used both subjective and devise-based methods to quantify PA.

3.3. Methodological quality of the included studies

Overall, we deemed only 13 studies (7%) to be of high quality, as they scored positively on all 5 or 4 of the quality items just described above. Ninety-two studies (49%) were of moderate quality, while 77 studies (41%) had low methodological quality. In 7 studies (4%), the overall methodological quality varied between low and moderate, or between moderate and high, depending on the domain and timeframe used for reporting results. Most studies did not report an adequate randomization procedure used to secure random allocation concealment, or they did not use blinded assessors to prevent information bias. A more detailed description of the risk of bias assessment of all items can be found in [S3 Table](#).

3.4. Effectiveness of school-based physical activity interventions

3.4.1. The short-term effect of interventions on changing children's PA level

Synthesis of the short-term intervention effects is based on PA data collected in 130,470 children/adolescents with a mean follow-up time of 51.4 weeks (minimum: 8 weeks; maximum 418 weeks) from baseline. Meta-analyses performed across the domains of children's total daily time, leisure time, and school time, are based on effect sizes reported in

122, 45, and 48 articles, respectively, with some articles reporting effect sizes across multiple domains. Ultimately, we used 287 different comparison groups (i.e., 155 different total daily-time effect sizes, 61 leisure time effect sizes, and 71 school-time effect sizes) for reporting the short-term intervention effect.

Overall, school-based PA interventions only had a small positive effect on children's and adolescents PA. Significant effect size interactions were observed across domains with the largest effect size being observed during school time (SMD = 0.37, 95% CI: 0.18; 0.55) followed by total daily time (SMD = 0.27, 95% CI: 0.19; 0.35), whereas no effect was observed during leisure time (SMD = 0.07, 95% CI: −0.03; 0.17). Large confidence intervals suggested considerable study variation in the observed effect sizes, although a significant modifying effect was observed due to domain. Consequently, high heterogeneity remained in all domain-specific results (i.e., school-time I^2 : 97.30%, total daily-time I^2 : 95.69%, leisure-time I^2 : 93.22%). The short-term intervention effect sizes are presented in [Fig. 2](#).

Subjective visual inspections of funnel plots drawn across PA assessment domains revealed asymmetric scatters for school time PA and total daily PA. Further, formal statistical testing by Egger's test supported small-study bias of the total daily PA intervention effect, and signs of small study bias for the school time PA intervention effect (Total day PA: $p < 0.0001$, School time PA: $p = 0.09$, leisure time PA: $p = 0.14$) ([Fig. 3](#)).

3.4.2. Effect sizes relative to intervention characteristics and study covariates

In total, we used 155 total-day intervention effects, reported in 122 articles, to map key factors potentially important for enhancing children's and adolescents PA level.

This meta-analysis revealed that the summary effect size might be larger in subgroup of interventions using a theoretical framework in the design phase (SMD = 0.33, 95% CI: 0.20; 0.45) compared to subgroup of interventions where no theoretical framework was applied (SMD = 0.20, 95% CI: 0.12; 0.28), although this was only borderline statistically significant ($p = 0.09$). No significant differences were observed in the summary effect size across subgroups of uni- and multi-component interventions (SMD 0.26 vs. 0.28; $p = 0.79$). The pooled analysis across subgroups of studies with and without parental involvement showed a statistically significant additional effect in favor of school-based PA interventions where parents were actively involved (SMD 0.35 vs. 0.16, $p = 0.02$). A significant positive intervention effect was observed when the intervention was being delivered by trained school staff (SMD = 0.21, 95% CI: 0.11; 0.32). The greatest effect was observed when the intervention was delivered by experts from outside school (SMD = 0.56, 95% CI: 0.13; 0.99) followed by combined delivery by experts and school staff (SMD = 0.28, 95% CI: 0.17; 0.39), although not statistically significant ($p = 0.14$) when comparing the summary effect size across all various modes of delivery. No significant intervention effect was observed when intervention delivery was by untrained school staff (SMD = 0.06, 95% CI: −0.16; 0.28) ([Fig. 4](#)). Generally, the intervention effect size differed tremendously across trials, and substantial heterogeneity remained also after considering the variability across relevant subgroups. When taking into account all study covariates in the same meta-regression model, τ^2 was reduced by 9.5% though (i.e., from 0.21 to 0.19).

3.4.3. The role of children's age and intervention program duration

No significant association between intervention program duration and change in children's and adolescents' total daily PA was observed ($\beta = -0.0009$, 95% CI: −0.0025; 0.0077) ([S1 Figure](#)). Children's and adolescents' age assessed at baseline, explained a very small part (0.96%) of the observed heterogeneity in results. The effect of the intervention on change in total daily PA increased by 0.03 (SMD) for each one-year increase in age, although only borderline significant and probably not clinically relevant (95% CI: −0.0037; 0.0697, $p = 0.078$). [S2 Figure](#)

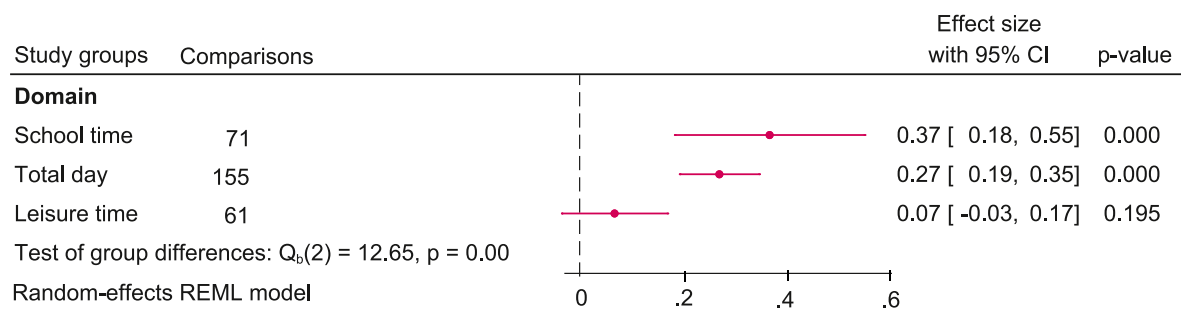


Fig. 2. Short-term intervention effect size across domains.

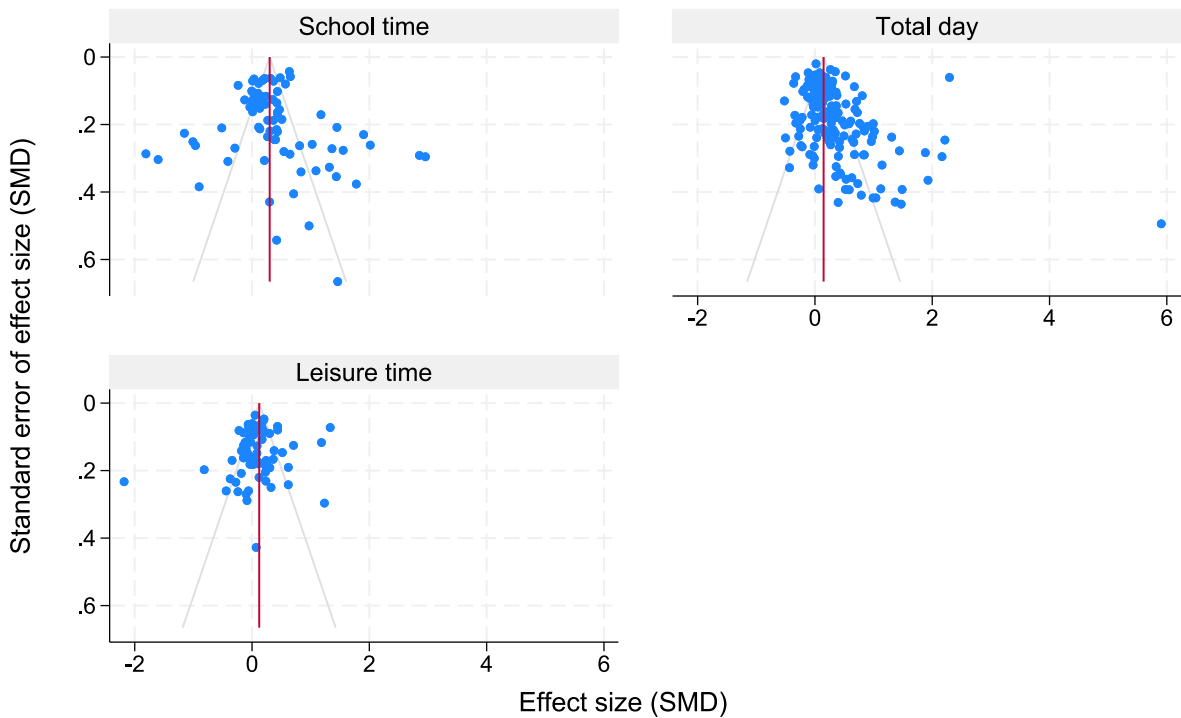


Fig. 3. Assessment of small-study short-term intervention effect across different assessment domains.

illustrates the relationship between participants’ age and summary effect size.

3.4.4. The long-term post intervention effect on children’s PA level

Synthesis of the long-term post intervention effect is bases on PA data collected in 17,819 children/adolescents with a mean-follow-up time of 48.4 weeks (minimum: 5 weeks; maximum 168 weeks) after end of the school-based program. The meta-analyses conducted across total daily time, leisure time, and school time are based on effect sizes reported in 27, 13, and 8 publications, respectively, and a total of 61 different comparison groups were utilized (i.e., 36 different total daily-time effect sizes, 16 leisure time effect sizes, and 9 school-time effect sizes) for reporting the long-term intervention effect.

Overall, school-based PA interventions had a small positive effect on children’s and adolescents PA level in a longer perspective after the intervention had ended. Borderline significant effect size interactions were observed across domains. The largest effect sizes were seen during children’s and adolescents’ school time (SMD = 0.35, 95% CI: 0.08; 0.62) and total daily time (SMD = 0.35, 95% CI: 0.19; 0.52), whereas no significant PA change was observed during leisure time (SMD = 0.12, 95% CI: -0.06; 0.30). Large I^2 value (i.e., school-time I^2 : 89.16%, total daily-time I^2 : 92.49%, and leisure-time I^2 : 89.16%) indicated substantial heterogeneity in the domain-specific intervention effects. The long-term

post intervention effect sizes are summarized in Fig. 5.

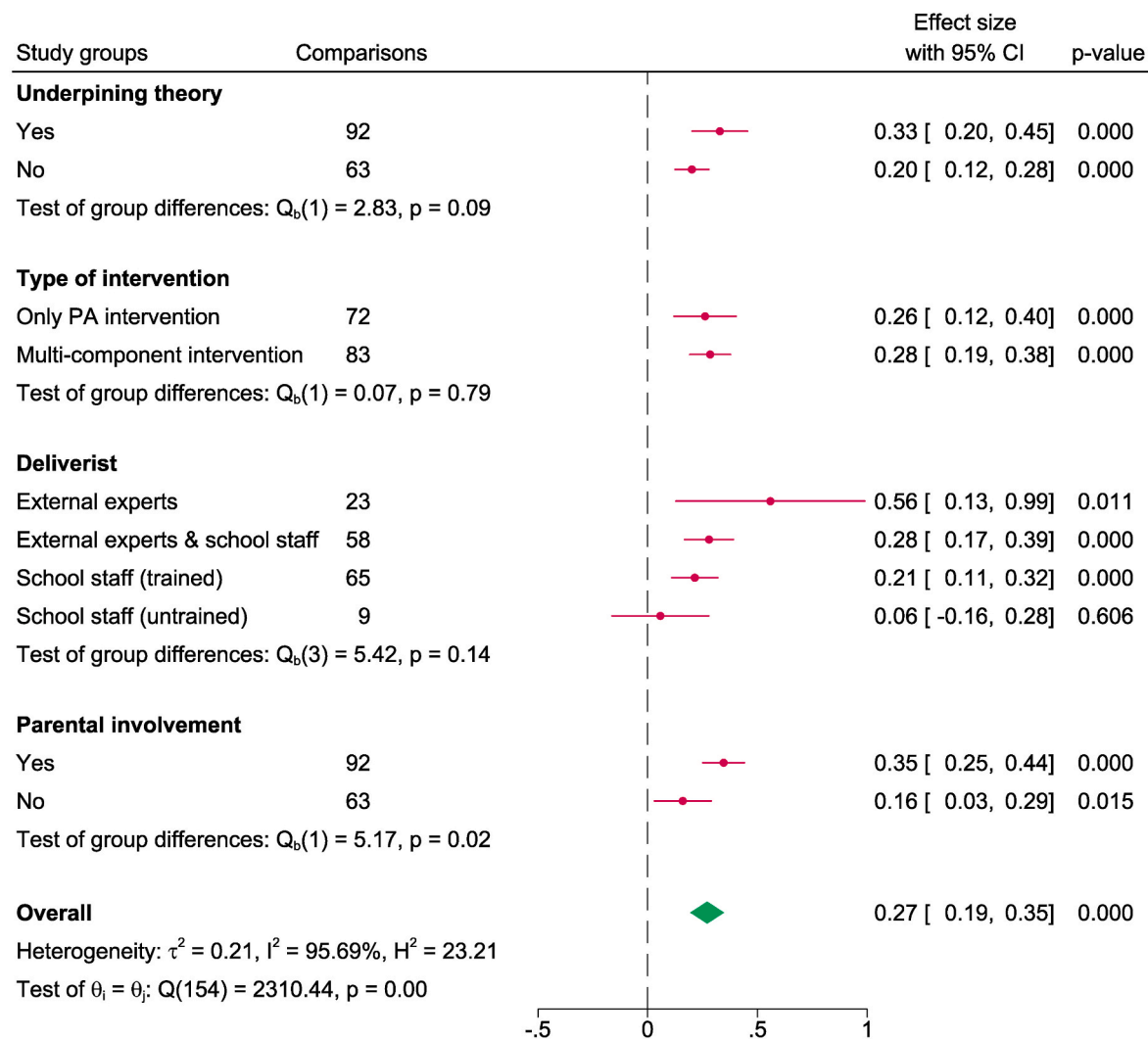
No significant association was observed between intervention duration and the post intervention long-term effect on children’s and adolescents’ total-day PA ($\beta = -0.002$, 95% CI: -0.0113; 0.0109) (S3 Figure).

Again, visual inspections of funnel plots (Fig. 6) showed asymmetric scatters for school-time PA and total daily PA, and small-study bias were supported by Egger’s test indicating small-study bias of school-time- and total-day PA intervention effects (school-time PA: $p = 0.0007$, total-day PA: $p < 0.0001$, leisure time PA: $p = 0.44$).

Summaries of the short- and the long-term intervention effectiveness are described in Table 1.

3.5. Sensitivity analyses

When statistical trim and fill simulations were used to eliminate small-study bias, the school-time- and total daily-time short-term intervention effects were reduced from SMD = 0.37, 95% CI: 0.19; 0.35 to SMD = 0.30, 95% CI: 0.27; 0.33, and from SMD = 0.27, 95% CI: 0.19; 0.35 to SMD = 0.07, 95% CI: 0.06; 0.09, respectively. In a similar manner, the long-term post intervention school-time PA- and total-day PA intervention effects were reduced from to SMD = 0.35, 95% CI: 0.08; 0.62 to SMD = 0.18, 95% CI: 0.11; 0.26, and from SMD = 0.35,



Random-effects REML model

Fig. 4. Short-term total-day intervention effects across a-priori selected program characteristics.

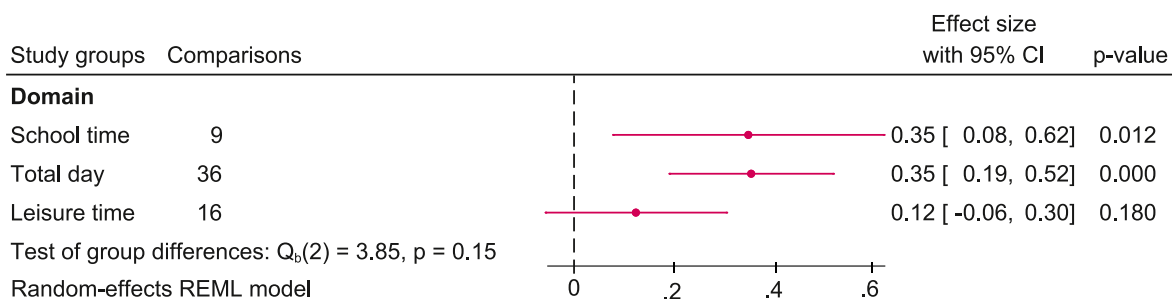


Fig. 5. Long-term intervention effect size across domains.

95% CI: 0.19; 0.52 to SMD = 0.06, 95% CI: 0.02; 0.10), respectively.

The meta-analysis results were only partly robust when exclusively including studies with high methodological quality, as the short-term overall daily summary intervention effect differed significantly ($p < 0.01$) across risk of bias status. More specifically, positive intervention effects were observed only in moderate- and high risk of bias studies (SMD = 0.35, 95% CI: 0.22; 0.49 and SMD = 0.22, 95% CI: 0.12; 0.32, respectively), whereas no intervention effect was observed in studies with low risk of bias (SMD = 0.05, 95% CI: -0.07; 0.17). No other

significant differences were observed in the summary effect sizes across subgroups of studies with varied level of quality, but the general tendency was that the effect size attenuated as risk of bias increased. Further, a tendency of a larger intervention effect was observed on children' and adolescents' total PA compared to their MVPA, especially regarding the long-term follow-up results. No consistent gender differences were observed, and the meta-analysis results were robust to subgroup analyses performed across literature search study phase, school stages, and PA construct and assessment methods, respectively. The

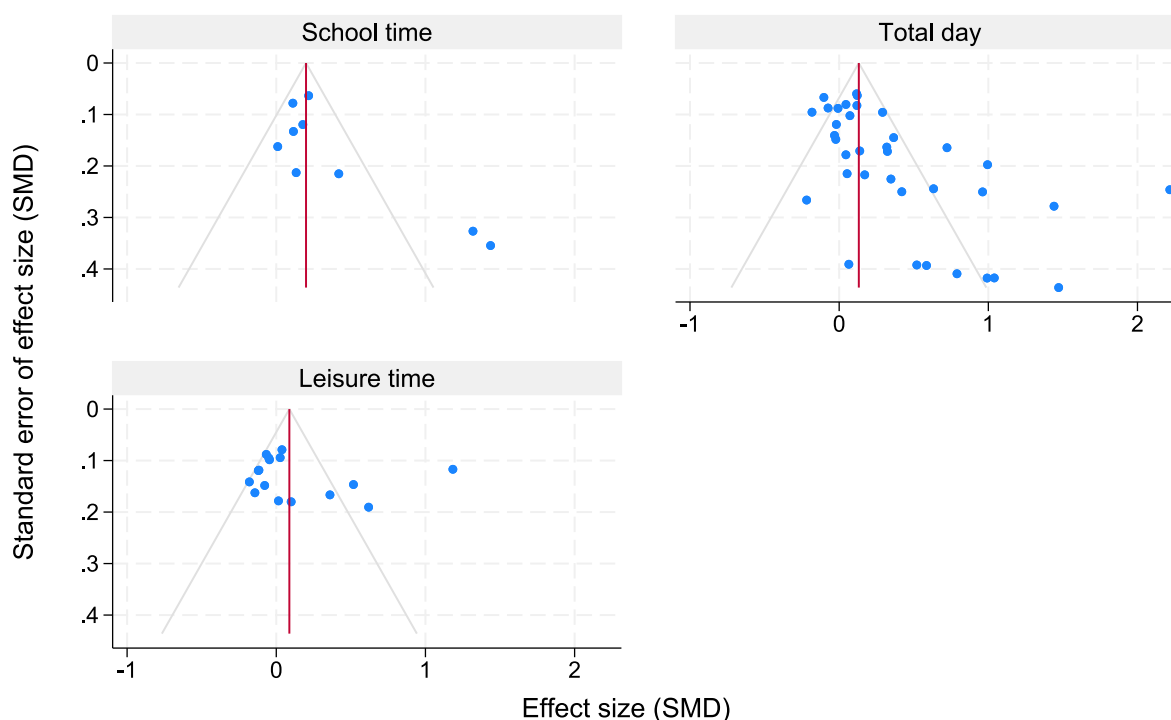


Fig. 6. Assessment of small-study long-term intervention effect across different assessment domains.

Table 1

Summary of findings.

Domain	Effect size in favor of intervention group (95% CI)	Number of participants (comparisons)	I^2 (Tau ²)	Certainty of the evidence (GRADE)
Short-term intervention effect				
School time	0.37 (0.18; 0.55)	21471 (71)	97.30 (0.57)	Low
Total day	0.27 (0.19; 0.35)	77057 (155)	95.69 (0.21)	Low
Leisure time	0.07 (−0.03; 0.17)	31942 (61)	93.22 (0.14)	Low
Long-term intervention effect				
School time	0.35 (0.08; 0.62)	2674 (9)	89.16 (0.14)	Low
Total day	0.35 (0.19; 0.52)	9561 (36)	92.49 (0.21)	Low
Leisure time	0.12 (−0.06; 0.30)	5584 (16)	89.16 (0.12)	Low

Effect sizes are reported in standardized mean differences (SMD).

results of all sensitivity analyses performed to explore possible sources of variation in the intervention effect size can be found in [S4 Table](#).

4. Discussion

To thoroughly synthesize the evidence of the effectiveness of school-based PA interventions aimed at increasing children's and adolescents' PA levels we collated standardized mean differences of overall- or moderate to vigorous PA assessed by means of self-reports or device-based methods during either total day, school time, or leisure time. We found that school-based PA interventions had a small but statistically significant positive effect on children's and adolescents' engagement in PA. Subgroup meta-analyses conducted across specific everyday domains revealed that interventions were more effective in increasing children's and adolescents' PA during school time compared to total daytime, and we found no intervention effects on their leisure-time PA level.

Our primary focus was on children's and adolescents' level of PA throughout the entire day, and the standardized short-term intervention effect observed during total daily time was 0.27 (SMD). Since meaningful interpretations of standardized differences can be complex from a clinical perspective, we post hoc converted this intervention effect into more easily understandable terms. By considering 22 min of MVPA per day as the standard deviation for differences between groups in total daily MVPA minutes from baseline to follow-up, which was based on the median from all studies included in our meta-analysis, we estimated that the observed intervention effect of 0.27 (SMD) would be equivalent to roughly 6 additional minutes of MVPA per day. While this might seem like a marginal increase, it appears relevant to consider whether the target population significantly deviates from meeting the PA recommendations or not, since even a small increase in daily MVPA minutes among those who are least active might have meaningful health benefits. No clear modifying effect of gender was observed. Moreover, we found no evidence of an association between intervention duration, school-stage, or participants' age, and intervention effect sizes. These findings were confirmed in additional post-hoc sensitivity analyses (data not shown) where the most extreme effect size (i.e., SMD = 5.9) observed in the study by [Lee et al. \(2012b\)](#) was excluded. Finally, small, but statistically significant, positive effect sizes in children's and adolescents' school-time- and total-day PA were confirmed also at long-term follow-up, and our findings indicate stronger sustainability of overall PA interventions effects compared to MVPA intervention effects.

It is possible that children compensate for increased PA at school by reducing PA in their leisure time. The ActivityStat hypothesis, which was first suggested by Rowland in 1998 ([Rowland, 1998](#)) proposes that an individual's biologically predefined PA level automatically will compensate for increased PA in one timeslot by reducing PA in another timeslot to preserve constant total energy expenditure. Other forms of compensation, e.g., if parents assume a smaller need for their child to participate in organized sports might also be a viable explanation. Also, if the administration of a school-based PA program results in a longer school-day it is plausible that participants reduce the time where they can engage in other forms of PA when outside school. This time-trade-off across different domains has been conceptualized by Olds and

colleagues (Olds et al., 2012) as the cross-elasticity of demands, meaning that time spent in one type of PA changes as the amount of time spent on another varies. Regardless of what might explain the lack of effect on children's leisure-time PA, and consequently the smaller effect observed in total-day PA, these findings should foster a debate about the relevance of initiating school-based interventions, especially if the aim is to increase children's health through promotion of full-day PA.

Like us, previous meta-analyses conducted to summarize the evidence of school-based interventions on children's and adolescents' PA have reported small positive effects in favor of the intervention group (Metcalfe et al., 2012; Neil-Sztramko et al., 2021; Nally et al., 2021), or no effects (Love et al., 2019; Borde et al., 2017). No increase, or a small increase at best, in children's and adolescents' PA from participating in school-based interventions might help explain e.g., the findings observed by Harris et al. (2009) who found no effect of school-based PA interventions on participants' body-mass index in a large meta-analysis including approximately 18,000 students aged 6–18 years old.

Mapping of key factors of particular importance for intervention effectiveness was investigated based on total-day PA effects. We observed significant interactions between parental involvement and effect size (i.e., interventions more effective if parents are involved in the program). Parental influence and the home environment have previously been shown to be factors impacting children's PA, and the family is generally perceived as an important source of inspiration and support related to children's behavior and lifestyle (Dowda et al., 2007). Some school-based interventions sometimes include PA homework tasks. Moreover, some interventions also include a component where parents attend activities at school. Consequently, parents are likely important in facilitating behavioral change in their children, also in the context of school-based PA interventions. The parents' role may be to guide children in the right direction towards desired behavioral change. This may be especially important in younger children. Our findings support the assumption that parents may be an important resource when the aim is to increase children's PA also through school-based interventions.

Training of personnel involved in the intervention delivery was the determining factor for program effectiveness, as no significant intervention effect was observed if program delivery was performed by schools' own untrained staff. Although not statistically significant, according to the persons who delivered the intervention (four categories) the largest effect sizes were observed when external experts delivered the program, followed by a delivery by both external experts and the schools' own staff, and then the school's own staff who had undergone training, respectively. The importance of external intervention delivery was previously highlighted in a systematic review examining the effect of school-based intervention on adolescents' PA and sedentary behavior (Hynynen et al., 2016). This indicates the importance of carefully considering how program delivery can be optimized when initiating interventions in the school setting to promote more active children.

We observed a borderline significant larger effect size in interventions where the planning and the content of the intervention program was based on a conceptual model supporting behavioral changes compared to those interventions that did not include such a conceptual model. This difference in effect size remained borderline significant, and only slightly attenuated, after we post-hoc tried to control for type of intervention (i.e., uni-vs. multi-component) (data not shown). We are aware of only one study, which has summarized the effect of school-based interventions on children's PA according to usage of an underlying theoretical framework in the program design phase. In a previous meta-analysis, Nally et al. found no evidence supporting a modifying effect on obesity-related outcomes based on whether a theory-based intervention approach was included (Nally et al., 2021). This questions the extent to which a theoretical model of behavioral change generally is useful when applied in various forms of complex school-based interventions. In our study, we noticed that many papers only stated that they used a behavior change model and merely mentioned the model's name. Thus, the papers mentioned nothing of

how the intervention was shaped from the theoretical model, which could suggest that the extent of theoretical foundation of the intervention in fact was minor.

Although pre-planned subgroup analyses generally explained some of the variation observed in the results across trials most of the variability remained largely unexplained. Consequently, the inconsistency is probably the result of large sources of variation in e.g., participants' backgrounds, trial design and methodology, school settings, and specific intervention program components utilized.

Subgroup analyses according to whether subjective self-reported or objective device-based PA assessments were used did not support effect size modifications due to the type of method used for quantifying participants' PA. In a previous meta-analysis summarizing the effect of school-based interventions on children and adolescents PA, Love et al. (2019) restricted their analyses to studies using accelerometer-assessed outcomes to foster a more homogeneous pool of trials due to poor validity and reliability of self-reported PA. Though accelerometers are generally considered a superior method for measuring PA in children and adolescents compared to PA measurements by self-report, there are still many sources of variations and errors to consider in accelerometer-based assessments, and, moreover, the ability of accelerometers to accurately assess children's PA energy expenditure during free living conditions has been questioned (Corder et al., 2007). Also, there is a lack of consensus regarding criteria for collecting and processing accelerometer data (Migueles et al., 2017), which hinders comparability across studies. Reanalysis of accelerometer data, as performed in the study by Love et al. (2019), will enhance such comparisons. Generally, adoptions of contemporary measurement- and data processing protocols, which are thought to be most accurate, have been requested to improve quality of methodology and results (Trost, 2020). In our study, we had an inclusive approach and found no evidence supporting that variability in intervention effects is reduced when applying only device-based outcomes when compared to self-reported outcomes.

Very few studies reported any data related to intervention fidelity, such as intervention adherence and quality of delivery. Without a more in-depth process evaluation it becomes more speculative to what degree participants were exposed to the intervention, and to what extent the intervention program was delivered as intended. It therefore remains a question to what degree the observed small intervention effect should be seen primarily in the light of the intervention program not being delivered as intended, participants only receiving a reduced intervention dose (i.e., low compliance), or a limited effectiveness of the intervention components. We therefore call for future school-based interventions to include process evaluations and report on these when main findings are reported.

We observed no differences in summary effect sizes according to whether the studies included uni-component interventions (i.e., programs only targeting changes in PA) or multi-component interventions (i.e., inclusion of program targeting both PA changes and changes in other health-related behavior such as diet). Findings from systematic reviews and meta-analyses comparing the effects of uni- and multi-component interventions on children's obesity and obesity-related behavior have been ambiguous, suggesting no differences (Katz et al., 2008), larger effect of single-component interventions (Nally et al., 2021), and slightly larger effect of multi-component programs (Liu et al., 2019). Such diverse findings might be due to trial specific differences in program outcomes, specific components included in programs, and a combination thereof. Only a small positive effect of multi-component interventions on children's daily PA level was confirmed in a meta-analysis by Russ et al. where a multi-component school-based PA intervention was conceptualized as programs including two or more of the following elements: physical education, PA before, during and after school hours, and family/community engagement (Russ et al., 2015). To be able to quantify the isolated effect of school-based PA interventions on children's and adolescents' PA level, we excluded trials that to a very

large extent included the local community or health care system as part of the intervention. We did, however, include interventions, which included the home setting as an intervention arena, typically in the form of PA homework, as we defined a priori parental involvement as a potential effect modifier.

Based on this systematic literature and meta-analysis we conclude that it is not possible to tease out the relative importance of specific PA intervention components for successful behavioral change. This conclusion is supported by the work conducted by others, e.g., [Liu et al. \(2019\)](#) who were not able to identify program components of multi-component interventions consistently associated with increased efficacy regarding change in children's and adolescents' improved body composition. Many school-based PA programs consist of a package of intervention components and the intervention to a large extent vary, e.g., regarding program development, content, and specific school setting or context in which the program is implemented. Consequently, no two interventions are identical. Therefore, it is not possible clearly to point out which intervention components should be the top priority when it comes to planning the most successful school-based intervention aimed at increasing children's PA. However, here, we identified a number of key factors indicating what drives effectiveness of complex school-based PA interventions.

Whether to use change scores or final scores as the primary outcome when conducting a meta-analysis is a statistical issue, which is subject for debate. In a randomized controlled trial, the mean difference from baseline to follow-up can usually be assumed to be the exact same underlying intervention effect as if final measurements were used for inclusion in the analyses. However, as we also included non-randomized controlled trials, the use of change scores to estimate the summarized effect sizes to some extent helped to account for group baseline differences, thereby strengthening the robustness of the observed results.

As a strength, this study synthesized the effect of school-based interventions on children's and adolescent's PA across multiple domains (i.e., school time, leisure time, and total time). In addition, a priori defined subgroup analyses were done to identify effect size modifications and relevant sources of variations. Another strength of our work was inclusion only of school-based intervention which assessed changes in PA outcomes and not health-related outcomes more broadly (e.g., changes in children's body composition or fitness level as proxies for a changed PA behavior).

A limitation of our study is that it was not possible for us to include trials published after September 2020 due to the extent of the literature and the enormous amount of resources used for extracting and harmonizing data. Also, the updated literature search (phase 2) was conducted using an upward citation search approach and not a systematic literature search. Considering the large number and diversity of the trials included in this study, we do not expect that our findings are biased due to unidentified literature. This is supported by sensitivity analyses comparing results observed in phase-1 and 2, respectively, in the present study.

In line with others ([van Sluijs et al., 2007](#); [Waters et al., 2011](#); [Hynynen et al., 2016](#)), we find that there is a strong need for more high-quality studies within the field of effect evaluation of school-based interventions on children's PA level, as the quality of most studies we included was only moderate or low. Generally, we observed that the effect sizes attenuated with reduced risk of bias (i.e., improved study quality). However, a statistically significant relationship between study quality and intervention effect size was observed only in the analysis of the short-term effect of changes in total daily PA.

The summary effect sizes reported in this study may be overestimated due to publication bias (small-study bias) as we observed an association between intervention effect size and number of participants enrolled in the trial. Statistical simulation, used to eliminate small-study bias, supported presence of publication bias, as simulated effect sizes were smaller when compared to un-simulated effect sizes. Although one should interpret simulated effect sizes with great caution, we could conform our overall conclusion when comparing the simulated and un-

simulated analyses, i.e., we identified a small but positive change in total-day PA and school-time PA in simulated meta-analyses.

We rated the overall certainty of evidence as low based on downgrading due to methodological study limitations, inconsistency of results, and small-study bias. Therefore, the small intervention effects observed, as well as the low or moderate quality observed in most included studies, together with the overall low-certainty evidence should be kept in mind when interpreting our findings. A more detailed description of the rating of certainty of evidence can be found in [S5 Table](#).

5. Conclusions

Our study demonstrates that school-based PA interventions only have a small positive effect on children's and adolescents' PA levels. Compared to the effects observed on their total daily PA, a slightly greater effect is observed on children's and adolescent's PA during school hours, while no intervention effect is observed during leisure time. However, interventions tend to be more effective if they make use of a theoretical behavior change framework in the program design phase. Training of persons involved in delivering the intervention is a determining factor for program effectiveness as no effect is observed if delivery is performed by schools' own untrained staff, and greater effect might be expected if external experts are utilized. Parents are an important resource also when it comes to school-based interventions, as the intervention effect is larger when parents are involved in the intervention program.

The small positive intervention effect is evident, also at long-term follow-up, as children and adolescents who participated in a school-based PA intervention are still more physically active during school time and during the whole day compared to children and adolescents who did not take part in an intervention. No clear modifying effect of age, gender or intervention duration is observed. Overall, the certainty of the evidence of the findings is rated as low.

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CRediT authorship contribution statement

Niels Christian Moeller: Writing – review & editing, Writing – original draft, Methodology, Funding acquisition, Formal analysis, Data curation, Conceptualization. **Lars Oestergaard:** Writing – review & editing, Methodology, Data curation, Conceptualization. **Martin Gillies Banke Rasmussen:** Writing – review & editing, Methodology, Data curation. **Jesper Schmidt-Persson:** Writing – review & editing, Methodology, Data curation. **Kristian Traberg Larsen:** Writing – review & editing, Data curation. **Carsten Bogh Juhl:** Writing – review & editing, Supervision, Methodology.

Declaration of competing interest

none

Data availability

I have shared the link to my data at the Attach file step.

[How to get children moving supporting data \(Original data\)](#) (Figshare)

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Appendix A. Supplementary data

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